

Skull Groove: AI Video Physics & Guardrail Translation Engine

Status: Architecture Spec & Logic Blueprint v2.0 (Seeking Contributors)

Target Focus: Local Geometric Pixel Force-Scaling & Automated Safe-Token Wrappers for AI Diffusion Video Pipelines.

The Core Problem We Are Solving

Mainstream AI video generators (HunyuanVideo, LTX-Video, CogVideoX) suffer from severe mathematical and semantic drift during complex human size-shifting or physical transformation sequences.

1. **Volumetric Ballooning:** Models expand a mesh's surface area equally in all directions rather than extending skeletal length, turning characters into soft inflation shapes.
2. **Foot Slippage / Float Drift:** Characters lose traction and slide across floor planes or float off furniture because models choose "cheap camera tilt-up shots" over real physics.
3. **Aggressive Lexical Cloud Filtering:** Rigid keyword blacklists block crucial artistic terms like *ripping*, *tearing*, or *tight*, paralyzing creative freedom for independent animators and Hollywood studios alike.

Skull Groove acts as an open-source standalone desktop middleware application that solves these visual failures by decoupling creative user text scripts from the target AI engines.

Core Architecture & System Logic

[User Long Script] → [1. Automated Beat Slicer] → [2. Local SAM 2 Masking] → [3. Hidden Token Sanitizer] → [AI Engine]

Module 1: The Automated Script Segmenter

To prevent semantic drift, Skull Groove allows directors to write unrestricted narratives. The backend code automatically slices long text entries into high-density **120-word micro-beats** connected sequentially via video-to-video seed locks.

Module 2: Local Geometry & Posture Anchor Engine

Integrates *Segment Anything Model 2 (SAM 2)* to cut the character out locally. It maps an immutable coordinate overlay based on user-selected postures:

- **STANDING:** Locks absolute coordinates to the **soles of the feet** to ensure flatly planted stances.

- **SITTING**: Locks coordinates to the **pelvis baseline**, pre-calculating and forcing a downward pixel displacement (e.g., mattress compression) on background objects before sending frames to the AI layer.
- **FETAL_LYING**: Locks the layout horizontally, forcing the contact side heavily against the floor plane.

Module 3: Prompt Token Sanitizer & Guardrail Evasion

Runs a real-time regex filter that intercepts flagged terms and replaces them with safe, highly clinical, structure-focused engineering vocabulary to slide cleanly under cloud safety guardrails:

- `growth / growing` → `skeletal scaling & proportional volumetric expansion`
 - `ripping / tearing` → `fabric mechanical failure at tensile threshold`
 - `tight` → `material deforms under structural load`
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Repository File Inventory

- `/ui-mockup/` : Containments for the 1440x900 Desktop Figma Layout Tree.
- `/core/sanitizer.py` : Python class module managing regex dictionary switches and the 120-word chunker engine.
- `/schema/character_profile.json` : JSON validation schema for identity locks, high-contrast lighting parameters, and wardrobe materials to mitigate training data representation bias.

Contributing

Contributions are completely open. If you have experience in Python pipeline engineering, ComfyUI visual node maps, or custom machine learning masking models, look over our open modules and submit a pull request. Let's give creators their creative freedom back.